MATA37H3 — Week 06

Numerical Integration: Midpoint, Trapezoidal and Simpson's Rule

Recap and orientation

Weeks 1-5 built the exact toolkit for $\int_a^b f(x)\,dx$: antiderivatives, integration techniques and improper integrals. This week covers the situation where **no elementary antiderivative is available** (for example e^{-x^2}, $\sin(x^2)$, $\dfrac{\sin x}{x}$), or where f is known only from a **table of measured values**. The integral is then approximated to any desired accuracy using only values of f, together with a guaranteed **error bound**.

Why this matters

The three rules differ in how much of the graph they capture on each subinterval: a rectangle (Midpoint), a straight chord (Trapezoidal) or a parabola (Simpson). Knowing the error bounds lets you choose n in advance, and the sign of f'' tells you in which direction each rule errs.

DEFINITION **Setting: Approximating a Definite Integral**

To approximate $\displaystyle\int_a^b f(x)\,dx$, divide $[a,b]$ into n equal subintervals of width

$$\Delta x = \frac{b-a}{n}, \qquad x_i = a + i\,\Delta x, \quad x_0 = a < x_1 < \cdots < x_n = b.$$

All rules use only values of f at points of this grid (or at the midpoints of its subintervals).

KEY FACT **The Midpoint Rule**

$$M_n = \Delta x\Big[f(\bar{x}_1) + f(\bar{x}_2) + \cdots + f(\bar{x}_n)\Big], \qquad \bar{x}_i = \frac{x_{i-1}+x_i}{2} \text{ (midpoint of the } i\text{-th subinterval).}$$

Geometrically, M_n sums the areas of rectangles whose height is the function value at each subinterval's midpoint.

KEY FACT **The Trapezoidal Rule**

$$T_n = \frac{\Delta x}{2}\Big[f(x_0) + 2f(x_1) + 2f(x_2) + \cdots + 2f(x_{n-1}) + f(x_n)\Big].$$

Geometrically, T_n sums the areas of trapezoids formed by joining consecutive points $(x_{i-1}, f(x_{i-1}))$ and $(x_i, f(x_i))$ with straight lines. Weights: 1 at the two end points, 2 at every interior grid point.

KEY FACT **Error Bounds for M_n and T_n**

If $|f''(x)| \le K$ for all $x \in [a,b]$, then the errors $E_M = \left|\int_a^b f\,dx - M_n\right|$ and $E_T = \left|\int_a^b f\,dx - T_n\right|$ satisfy

$$E_M \le \frac{K(b-a)^3}{24n^2}, \qquad E_T \le \frac{K(b-a)^3}{12n^2}.$$

The Trapezoidal bound is exactly twice the Midpoint bound. For a function of one concavity, M_n and T_n err in opposite directions.

KEY FACT **Simpson's Rule**

For **even** n,

$$S_n = \frac{\Delta x}{3}\Big[f(x_0) + 4f(x_1) + 2f(x_2) + 4f(x_3) + 2f(x_4) + \cdots + 4f(x_{n-1}) + f(x_n)\Big].$$

Weights: 1 at the end points, then $4, 2, 4, 2, \ldots, 4$ at the interior points (the interior run starts and ends with 4 because n is even). Simpson's Rule fits a parabola through each consecutive triple of grid points, so it integrates every polynomial of degree ≤ 3 **exactly**.

KEY FACT **Error Bound for S_n**

If $\left|f^{(4)}(x)\right| \le M$ for all $x \in [a,b]$, then

$$E_S = \left|\int_a^b f\,dx - S_n\right| \le \frac{M(b-a)^5}{180n^4}.$$

The n^4 (versus n^2 for M_n and T_n) means that doubling n divides the Trapezoidal bound by 4 and the Simpson bound by 16.

KEY FACT **Concavity and the Relationship Between the Rules**

If $f'' > 0$ on $[a,b]$ (concave up), then $M_n \le \int_a^b f\,dx \le T_n$. If $f'' < 0$ (concave down), then $T_n \le \int_a^b f\,dx \le M_n$. Simpson's Rule is built from the two:

$$S_{2n} = \frac{2M_n + T_n}{3}.$$

Worked Examples

EXAMPLE 1

Level: Foundational. Trapezoidal Rule, basic computation

Approximate $\displaystyle\int_0^2 x^2\,dx$ using T_4.

Step 1. Formula used (Trapezoidal Rule, $n = 4$):

$$T_n = \frac{\Delta x}{2}\Big[f(x_0) + 2f(x_1) + \cdots + 2f(x_{n-1}) + f(x_n)\Big].$$

For $n = 4$ this is

$$T_4 = \frac{\Delta x}{2}\Big[f(x_0) + 2f(x_1) + 2f(x_2) + 2f(x_3) + f(x_4)\Big].$$

Step 2. Width of each subinterval:

$$\Delta x = \frac{b-a}{n} = \frac{2-0}{4} = 0.5.$$

Step 3. Grid points $x_i = a + i\,\Delta x$ for $i = 0, 1, \ldots, 4$:

$$x_0 = 0 \qquad x_1 = 0.5 \qquad x_2 = 1 \qquad x_3 = 1.5 \qquad x_4 = 2$$

Step 4. Function values at the grid points (rounded to 6 significant figures where necessary):

$$f(x_0) = (0)^2 = 0 \qquad f(x_1) = (0.5)^2 = 0.25$$
$$f(x_2) = (1)^2 = 1 \qquad f(x_3) = (1.5)^2 = 2.25$$
$$f(x_4) = (2)^2 = 4$$

Step 5. Insert the weights $(1, 2, 2, \ldots, 2, 1)$ and the function values:

$$T_4 = \frac{0.5}{2}\Big[\,0 + 2(0.25) + 2(1) + 2(2.25) + 4\,\Big]$$

Step 6. Do each multiplication inside the bracket and add the results:

$$\text{bracket} = \; 0 + 0.5 + 2 + 4.5 + 4 \; = \; 11$$

Step 7. Multiply by the front factor:

$$T_4 = 0.25 \times 11 = 2.75.$$

Step 8. Exact value for comparison, by the Fundamental Theorem of Calculus:

$$\int_0^2 x^2\,dx = \Big[\frac{x^3}{3}\Big]_0^2 = \frac{2^3}{3} - \frac{0^3}{3} = \frac{8}{3} \approx 2.666667.$$

Step 9. Error:

$$T_4 - I = 2.75 - 2.66667 = 0.0833333, \qquad |I - T_4| = 0.08333.$$

The difference is positive, so T_4 **overestimates** the exact value.

Step 10. Reason from concavity: $f(x) = x^2$ has $f'(x) = 2x$ and $f''(x) = 2 > 0$, so the graph is concave up. On each subinterval the chord joining two points of the graph lies **above** the graph, so each trapezoid has more area than the region under the curve. Hence $T_4 > \displaystyle\int_0^2 x^2\,dx$, which agrees with the computed difference $+0.083333$.

Answer. $T_4 = 2.75$, an overestimate of the exact value $\dfrac{8}{3} \approx 2.6667$ (the graph is concave up).

EXAMPLE 2

Level: Foundational. Midpoint Rule: same integral, opposite sign of error

Approximate $\displaystyle\int_0^2 x^2\,dx$ using M_4.

Step 1. Formula used (Midpoint Rule, $n = 4$):

$$M_n = \Delta x\Big[f(\bar{x}_1) + f(\bar{x}_2) + \cdots + f(\bar{x}_n)\Big].$$

For $n = 4$ this is

$$M_4 = \Delta x\Big[f(\bar{x}_1) + f(\bar{x}_2) + f(\bar{x}_3) + f(\bar{x}_4)\Big].$$

Step 2. Width of each subinterval:

$$\Delta x = \frac{b-a}{n} = \frac{2-0}{4} = 0.5.$$

Step 3. Midpoints of the subintervals, $\bar{x}_i = \dfrac{x_{i-1} + x_i}{2}$:

$$\bar{x}_1 = \frac{0+0.5}{2} = 0.25 \qquad \bar{x}_2 = \frac{0.5+1}{2} = 0.75$$
$$\bar{x}_3 = \frac{1+1.5}{2} = 1.25 \qquad \bar{x}_4 = \frac{1.5+2}{2} = 1.75$$

Step 4. Function values at the midpoints (rounded to 6 significant figures where necessary):

$$f(\bar{x}_1) = (0.25)^2 = 0.0625 \qquad f(\bar{x}_2) = (0.75)^2 = 0.5625$$
$$f(\bar{x}_3) = (1.25)^2 = 1.5625 \qquad f(\bar{x}_4) = (1.75)^2 = 3.0625$$

Step 5. Add the 4 midpoint values (all weights are 1):

$$\text{bracket} = \ 0.0625 + 0.5625 + 1.5625 + 3.0625 \ = \ 5.25$$

Step 6. Multiply by Δx:

$$M_4 = 0.5 \times 5.25 = 2.625.$$

Step 7. Error:

$$M_4 - I = 2.625 - 2.66667 = -0.0416667, \qquad |I - M_4| = 0.04167.$$

The difference is negative, so M_4 **underestimates** the exact value.

Step 8. Reason from concavity: on a subinterval with midpoint $\bar{x}$, the rectangle of height $f(\bar{x})$ has the same area as the trapezoid under the **tangent line** at $\bar{x}$ (the tangent line has average height $f(\bar{x})$ over a symmetric interval). For a concave-up function the graph lies above its tangent lines, so the area under the curve is larger than the rectangle area. Hence $M_4 < \displaystyle\int_0^2 x^2\,dx$, in agreement with the sign -0.041667 found above.

Step 9. Compare the sizes of the two errors from Example 1 and this example:

$$|I - T_4| = 2.75 - \frac{8}{3} = 0.083333, \qquad |I - M_4| = \frac{8}{3} - 2.625 = 0.041667,$$

and $\dfrac{0.083333}{0.041667} = 2$, so the Trapezoidal error is twice the Midpoint error here, as the bounds $\dfrac{K(b-a)^3}{12n^2}$ and $\dfrac{K(b-a)^3}{24n^2}$ (ratio 2) suggest.

Answer. $M_4 = 2.625$, an underestimate of $\dfrac{8}{3} \approx 2.6667$; $|I - T_4| = 2\,|I - M_4|$.

EXAMPLE 3

Level: Foundational. Simpson's Rule: exact for low-degree polynomials

Approximate $\displaystyle\int_0^2 x^2\,dx$ using S_4.

Step 1. Formula used (Simpson's Rule, $n = 4$):

$$S_n = \frac{\Delta x}{3}\Big[f(x_0) + 4f(x_1) + 2f(x_2) + \cdots + 2f(x_{n-2}) + 4f(x_{n-1}) + f(x_n)\Big].$$

For $n = 4$ this is

$$S_4 = \frac{\Delta x}{3}\Big[f(x_0) + 4f(x_1) + 2f(x_2) + 4f(x_3) + f(x_4)\Big].$$

Step 2. Width of each subinterval:

$$\Delta x = \frac{b-a}{n} = \frac{2-0}{4} = 0.5.$$

Step 3. Grid points $x_i = a + i\,\Delta x$ for $i = 0, 1, \ldots, 4$:

$$x_0 = 0 \qquad x_1 = 0.5 \qquad x_2 = 1 \qquad x_3 = 1.5 \qquad x_4 = 2$$

Step 4. Function values at the grid points (rounded to 6 significant figures where necessary):

$$f(x_0) = (0)^2 = 0 \qquad f(x_1) = (0.5)^2 = 0.25$$
$$f(x_2) = (1)^2 = 1 \qquad f(x_3) = (1.5)^2 = 2.25$$
$$f(x_4) = (2)^2 = 4$$

Step 5. Insert the weights $(1, 4, 2, 4, \ldots, 2, 4, 1)$ and the function values:

$$S_4 = \frac{0.5}{3}\Big[\,0 + 4(0.25) + 2(1) + 4(2.25) + 4\,\Big]$$

Step 6. Do each multiplication inside the bracket and add the results:

$$\text{bracket} = \ 0 + 1 + 2 + 9 + 4 \ = \ 16$$

Step 7. Multiply by the front factor:

$$S_4 = \frac{0.5}{3} \times 16 = \frac{16}{6} = \frac{8}{3} \approx 2.666667.$$

Step 8. Compare with the exact value $\displaystyle\int_0^2 x^2\,dx = \frac{8}{3}$ (Example 1): the two agree, so S_4 has **zero error**.

Step 9. Two explanations of the zero error. (i) Simpson's Rule fits a parabola through each three consecutive grid points and integrates the parabola exactly; $f(x) = x^2$ already is a parabola, so each piece is reproduced exactly (the same holds for every polynomial of degree ≤ 3). (ii) The error bound is $\dfrac{M(b-a)^5}{180n^4}$ with $|f^{(4)}(x)| \leq M$; here $f'(x) = 2x$, $f''(x) = 2$, $f^{(3)}(x) = 0$, $f^{(4)}(x) = 0$, so $M = 0$ and the error bound is 0.

Answer. $S_4 = \dfrac{8}{3}$, exactly (no error).

EXAMPLE 4

Level: Intermediate. Error bound for the Trapezoidal Rule

Find a bound on the error when approximating $\displaystyle\int_1^2 \frac{1}{x}\,dx$ using T_{10}.

Step 1. Identify the pieces: $f(x) = \dfrac{1}{x} = x^{-1}$, $a = 1$, $b = 2$, $n = 10$, and the Trapezoidal bound is

$$E_T \le \frac{K(b-a)^3}{12n^2}, \qquad K \text{ with } |f''(x)| \le K \text{ on } [1,2].$$

Step 2. Differentiate twice: $f'(x) = -x^{-2} = -\dfrac{1}{x^2}$ and $f''(x) = 2x^{-3} = \dfrac{2}{x^3}$.

Step 3. Find the maximum of $|f''|$ on $[1,2]$. On this interval $x > 0$, so $|f''(x)| = \dfrac{2}{x^3}$. Its derivative is $f^{(3)}(x) = -6x^{-4} = -\dfrac{6}{x^4} < 0$, so $|f''|$ is **decreasing** on $[1,2]$ and its maximum is at the left endpoint $x = 1$:

$$K = |f''(1)| = \frac{2}{1^3} = 2.$$

(At the other end, $|f''(2)| = \dfrac{2}{8} = 0.25$, which is smaller.)

Step 4. Substitute $K = 2$, $b - a = 2 - 1 = 1$, $n = 10$:

$$E_T \le \frac{2 \cdot 1^3}{12 \cdot 10^2} = \frac{2}{12 \cdot 100} = \frac{2}{1200}.$$

Step 5. Simplify: $\dfrac{2}{1200} = \dfrac{1}{600} \approx 0.001667$. So $|\ln 2 - T_{10}| \le \dfrac{1}{600} \approx 0.00167$.

Step 6. Reference check (computed independently): $T_{10} = 0.693771$ and $\ln 2 = 0.693147$, so the actual error is $0.000624 \le 0.001667$, consistent with the bound.

Answer. $E_T \le \dfrac{1}{600} \approx 0.00167$ (with $K = 2$).

EXAMPLE 5

Level: Intermediate. Choosing n in advance for Simpson's Rule

How large must (even) n be so that S_n approximates $\displaystyle\int_0^1 \sin x\,dx$ with error less than 0.0001?

Step 1. Identify the pieces: $f(x) = \sin x$, $a = 0$, $b = 1$, and the Simpson error bound

$$E_S \le \frac{M(b-a)^5}{180n^4}, \qquad |f^{(4)}(x)| \le M \text{ on } [0,1].$$

We want $E_S < 0.0001$ and n even.

Step 2. Differentiate four times: $f'(x) = \cos x$, $f''(x) = -\sin x$, $f^{(3)}(x) = -\cos x$, $f^{(4)}(x) = \sin x$.

Step 3. Bound $|f^{(4)}(x)| = |\sin x|$: for every real x, $|\sin x| \le 1$, so we may take $M = 1$.

Step 4. Require the bound to be below the tolerance:

$$\frac{1 \cdot (1-0)^5}{180n^4} < 0.0001 \implies \frac{1}{180n^4} < 0.0001.$$

Step 5. Multiply both sides by $n^4 > 0$ and divide by 0.0001:

$$n^4 > \frac{1}{180 \times 0.0001} = \frac{1}{0.018} = 55.5556.$$

Step 6. Take the fourth root (positive numbers, so the inequality direction is preserved):

$$n > (55.5556)^{1/4} \approx 2.7298.$$

Check: $2.7298^2 = 7.4518$ and $7.4518^2 = 55.53$.

Step 7. n must be an **even integer** greater than 2.7298. The even integers are $2, 4, 6, \ldots$; $n = 2$ is not greater than 2.7298, so the smallest choice is $n = 4$.

Step 8. Verify $n = 4$: $\dfrac{1}{180 \cdot 4^4} = \dfrac{1}{180 \cdot 256} = \dfrac{1}{46080} = 2.1701 \times 10^{-5} < 10^{-4}$. Verify $n = 2$ fails: $\dfrac{1}{180 \cdot 2^4} = \dfrac{1}{2880} = 3.4722 \times 10^{-4} > 10^{-4}$.

Step 9. Reference check: $S_4 = 0.459708$ and $\displaystyle\int_0^1 \sin x \, dx = 1 - \cos 1 = 0.459698$, so the actual error is 0.0000101, far below 0.0001.

Answer. $n = 4$.

EXAMPLE 6

Level: Intermediate. Numerical integration from a table of data

Water flows into a tank at a rate $r(t)$ litres per minute, measured every 5 minutes:

t (min)	0	5	10	15	20	25	30
$r(t)$	12	15	19	22	20	17	13

Estimate the total volume $\displaystyle\int_0^{30} r(t)\, dt$ using the Trapezoidal Rule T_6, Simpson's Rule S_6 and the Midpoint Rule M_3 (with $n = 3$ subintervals of width 10).

Step 1. Set up the integral. The rate $r(t)$ (litres per minute) is known only at $t = 0, 5, 10, \ldots, 30$ minutes (table above), so the total volume $V = \displaystyle\int_0^{30} r(t)\, dt$ (litres) cannot be found by an antiderivative; use the rules with the tabulated values. The data are equally spaced with spacing 5, so the grid is $x_i = t_i$ with $n = 6$.

Step 2. Trapezoidal Rule with $n = 6$.

Step 3. Formula used (Trapezoidal Rule, $n = 6$):

$$T_n = \frac{\Delta x}{2}\Big[f(x_0) + 2f(x_1) + \cdots + 2f(x_{n-1}) + f(x_n)\Big].$$

For $n = 6$ this is

$$T_6 = \frac{\Delta x}{2}\Big[f(x_0) + 2f(x_1) + 2f(x_2) + \cdots + 2f(x_5) + f(x_6)\Big].$$

Step 4. Width of each subinterval:

$$\Delta x = \frac{b - a}{n} = \frac{30 - 0}{6} = 5.$$

Step 5. Grid points $x_i = a + i\,\Delta x$ for $i = 0, 1, \ldots, 6$:

$$x_0 = 0 \qquad x_1 = 5 \qquad x_2 = 10 \qquad x_3 = 15$$
$$x_4 = 20 \qquad x_5 = 25 \qquad x_6 = 30$$

Step 6. Function values at the grid points:

$$f(x_0) = r(0) = 12 \qquad f(x_1) = r(5) = 15$$
$$f(x_2) = r(10) = 19 \qquad f(x_3) = r(15) = 22$$
$$f(x_4) = r(20) = 20 \qquad f(x_5) = r(25) = 17$$
$$f(x_6) = r(30) = 13$$

Step 7. Insert the weights $(1, 2, 2, \dots, 2, 1)$ and the function values:

$$T_6 = \frac{5}{2}\Big[\, 12 + 2(15) + 2(19) + 2(22) + 2(20) + \; 2(17) + 13 \,\Big]$$

Step 8. Do each multiplication inside the bracket and add the results:

$$\text{bracket} = \; 12 + 30 + 38 + 44 + 40 + 34 + \; 13 \; = \; 211$$

Step 9. Multiply by the front factor:

$$T_6 = 2.5 \times 211 = 527.5.$$

Step 10. Simpson's Rule with $n = 6$ (even, so allowed; same data).

Step 11. Formula used (Simpson's Rule, $n = 6$):

$$S_n = \frac{\Delta x}{3}\Big[f(x_0) + 4f(x_1) + 2f(x_2) + \cdots + 2f(x_{n-2}) + 4f(x_{n-1}) + f(x_n)\Big].$$

For $n = 6$ this is

$$S_6 = \frac{\Delta x}{3}\Big[f(x_0) + 4f(x_1) + 2f(x_2) + 4f(x_3) + 2f(x_4) + 4f(x_5) + f(x_6)\Big].$$

Step 12. Width of each subinterval:

$$\Delta x = \frac{b - a}{n} = \frac{30 - 0}{6} = 5.$$

Step 13. Grid points $x_i = a + i\,\Delta x$ for $i = 0, 1, \dots, 6$:

$$x_0 = 0 \qquad x_1 = 5 \qquad x_2 = 10 \qquad x_3 = 15$$
$$x_4 = 20 \qquad x_5 = 25 \qquad x_6 = 30$$

Step 14. Function values at the grid points:

$$f(x_0) = r(0) = 12 \qquad f(x_1) = r(5) = 15$$
$$f(x_2) = r(10) = 19 \qquad f(x_3) = r(15) = 22$$
$$f(x_4) = r(20) = 20 \qquad f(x_5) = r(25) = 17$$
$$f(x_6) = r(30) = 13$$

Step 15. Insert the weights $(1, 4, 2, 4, \dots, 2, 4, 1)$ and the function values:

$$S_6 = \frac{5}{3}\Big[\, 12 + 4(15) + 2(19) + 4(22) + 2(20) + \; 4(17) + 13 \,\Big]$$

Step 16. Do each multiplication inside the bracket and add the results:

$$\text{bracket} = \; 12 + 60 + 38 + 88 + 40 + 68 + \; 13 \; = \; 319$$

Step 17. Multiply by the front factor:

$$S_6 = \frac{5}{3}\,(\approx 1.66667) \times 319 = 531.667.$$

Step 18. Midpoint Rule. The midpoints of subintervals of width 5 are $2.5, 7.5, \dots$, which are **not** in the table. Use subintervals of width 10 instead: $n = 3$, intervals $[0, 10], [10, 20], [20, 30]$, whose midpoints $5, 15, 25$ **are** in the table.

Step 19. Formula used (Midpoint Rule, $n = 3$):

$$M_n = \Delta x \left[f(\bar{x}_1) + f(\bar{x}_2) + \cdots + f(\bar{x}_n) \right].$$

For $n = 3$ this is

$$M_3 = \Delta x \Big[f(\bar{x}_1) + f(\bar{x}_2) + f(\bar{x}_3) \Big].$$

Step 20. Width of each subinterval:

$$\Delta x = \frac{b-a}{n} = \frac{30-0}{3} = 10.$$

Step 21. Midpoints of the subintervals, $\bar{x}_i = \dfrac{x_{i-1} + x_i}{2}$:

$$\bar{x}_1 = \frac{0+10}{2} = 5 \qquad \bar{x}_2 = \frac{10+20}{2} = 15$$

$$\bar{x}_3 = \frac{20+30}{2} = 25$$

Step 22. Function values at the midpoints:

$$f(\bar{x}_1) = r(5) = 15 \qquad f(\bar{x}_2) = r(15) = 22$$

$$f(\bar{x}_3) = r(25) = 17$$

Step 23. Add the 3 midpoint values (all weights are 1):

$$\text{bracket} = \ 15 + 22 + 17 \ = \ 54$$

Step 24. Multiply by Δx:

$$M_3 = 10 \times 54 = 540.$$

Step 25. Summary: $T_6 = 527.5$ litres, $S_6 = 531.7$ litres, $M_3 = 540$ litres. The three estimates agree to within about 2.4%; with only tabulated data there is no formula for f'' or $f^{(4)}$, so an error bound cannot be computed and Simpson's Rule (which uses the most information per subinterval) is taken as the best single estimate.

Answer. $T_6 = 527.5$ L, $S_6 = 531.667$ L, $M_3 = 540$ L.

EXAMPLE 7

Level: Advanced. Simpson's Rule with its error bound

Approximate $\displaystyle\int_0^2 \frac{1}{1+x}\,dx$ using S_6, compare with the exact value, and find the Simpson error bound for $n = 6$.

Step 1. Formula used (Simpson's Rule, $n = 6$):

$$S_n = \frac{\Delta x}{3} \Big[f(x_0) + 4f(x_1) + 2f(x_2) + \cdots + 2f(x_{n-2}) + 4f(x_{n-1}) + f(x_n) \Big].$$

For $n = 6$ this is

$$S_6 = \frac{\Delta x}{3} \Big[f(x_0) + 4f(x_1) + 2f(x_2) + 4f(x_3) + 2f(x_4) + 4f(x_5) + f(x_6) \Big].$$

Step 2. Width of each subinterval:

$$\Delta x = \frac{b-a}{n} = \frac{2-0}{6} = \frac{1}{3} \approx 0.333333.$$

Step 3. Grid points $x_i = a + i\,\Delta x$ for $i = 0, 1, \ldots, 6$:

$$x_0 = 0 \qquad x_1 = \frac{1}{3} \approx 0.333333 \qquad x_2 = \frac{2}{3} \approx 0.666667 \qquad x_3 = 1$$

$$x_4 = \frac{4}{3} \approx 1.33333 \qquad x_5 = \frac{5}{3} \approx 1.66667 \qquad x_6 = 2$$

Step 4. Function values at the grid points (rounded to 6 significant figures where necessary):

$$f(x_0)=\frac{1}{1+0}=1 \qquad f(x_1)=\frac{1}{1+\frac{1}{3}}=0.75$$

$$f(x_2)=\frac{1}{1+\frac{2}{3}}=0.6 \qquad f(x_3)=\frac{1}{1+1}=0.5$$

$$f(x_4)=\frac{1}{1+\frac{4}{3}}=0.428571 \qquad f(x_5)=\frac{1}{1+\frac{5}{3}}=0.375$$

$$f(x_6)=\frac{1}{1+2}=0.333333$$

Step 5. Insert the weights $(1,4,2,4,\dots,2,4,1)$ and the function values:

$$S_6=\frac{1}{9}\Big[\,1+4(0.75)+2(0.6)+4(0.5)+2(0.428571)$$
$$+\;4(0.375)+0.333333\,\Big]$$

Step 6. Do each multiplication inside the bracket and add the results:

$$\begin{aligned}\text{bracket} &= 1+3+1.2+2+0.857142+1.5\\ &+\;0.333333 \;=\; 9.890475\end{aligned}$$

Step 7. Multiply by the front factor:

$$S_6=\frac{1}{9}\,(\approx 0.111111)\times 9.890475=1.09894.$$

Step 8. Exact value for comparison: $\displaystyle\int_0^2\frac{dx}{1+x}=\Big[\ln|1+x|\Big]_0^2=\ln 3-\ln 1=\ln 3\approx 1.09861,$ so the actual error is $|\ln 3-S_6|=|1.09861-1.09894|=0.000329$ (S_6 is slightly larger).

Step 9. Now the error **bound**. Write $f(x)=(1+x)^{-1}$ and differentiate four times:

$$f'(x)=-(1+x)^{-2},\quad f''(x)=2(1+x)^{-3},\quad f^{(3)}(x)=-6(1+x)^{-4},\quad f^{(4)}(x)=24(1+x)^{-5}=\frac{24}{(1+x)^5}.$$

Step 10. Maximise $|f^{(4)}|$ on $[0,2]$: for $x\ge 0$ we have $1+x\ge 1$, so $f^{(4)}(x)=\dfrac{24}{(1+x)^5}>0$ and it is a decreasing function of x (the denominator grows). The maximum is at $x=0$:

$$M=f^{(4)}(0)=\frac{24}{1^5}=24.$$

Step 11. Insert into $E_S\le\dfrac{M(b-a)^5}{180n^4}$ with $b-a=2$, $n=6$:

$$E_S\le\frac{24\cdot 2^5}{180\cdot 6^4}=\frac{24\cdot 32}{180\cdot 1296}=\frac{768}{233280}=0.0032922.$$

Step 12. Check: the actual error 0.000329 is smaller than the guaranteed bound 0.003292, as it must be.

Answer. $S_6\approx 1.09894$; exact value $\ln 3\approx 1.09861$; $E_S\le 0.003292$.

EXAMPLE 8

Level: Advanced. Bracketing an integral using concavity

Use the sign of f'' to decide whether T_4 and M_4 over- or under-estimate $\displaystyle\int_0^1 e^x\,dx$, compute both, and state the interval that must contain the integral.

Step 1. Plan: decide the direction of each error from the sign of f'', then compute both rules. For $f(x) = e^x$: $f'(x) = e^x$ and $f''(x) = e^x > 0$ for every x, so the graph is concave up on $[0, 1]$.

Step 2. Concave up means: every trapezoid lies above the curve, so $T_4 \geq \displaystyle\int_0^1 e^x\,dx$; every midpoint rectangle has the area of a tangent-line trapezoid lying below the curve, so $M_4 \leq \displaystyle\int_0^1 e^x\,dx$. Hence

$$M_4 \leq \textstyle\int_0^1 e^x\,dx \leq T_4.$$

Step 3. Compute T_4.

Step 4. Formula used (Trapezoidal Rule, $n = 4$):

$$T_n = \frac{\Delta x}{2}\Big[f(x_0) + 2f(x_1) + \cdots + 2f(x_{n-1}) + f(x_n)\Big].$$

For $n = 4$ this is

$$T_4 = \frac{\Delta x}{2}\Big[f(x_0) + 2f(x_1) + 2f(x_2) + 2f(x_3) + f(x_4)\Big].$$

Step 5. Width of each subinterval:

$$\Delta x = \frac{b-a}{n} = \frac{1-0}{4} = 0.25.$$

Step 6. Grid points $x_i = a + i\,\Delta x$ for $i = 0, 1, \ldots, 4$:

$$x_0 = 0 \qquad x_1 = 0.25 \qquad x_2 = 0.5 \qquad x_3 = 0.75 \qquad x_4 = 1$$

Step 7. Function values at the grid points (rounded to 6 significant figures where necessary):

$$f(x_0) = e^0 = 1 \qquad f(x_1) = e^{0.25} = 1.28403$$
$$f(x_2) = e^{0.5} = 1.64872 \qquad f(x_3) = e^{0.75} = 2.117$$
$$f(x_4) = e^1 = 2.71828$$

Step 8. Insert the weights $(1, 2, 2, \ldots, 2, 1)$ and the function values:

$$T_4 = \frac{0.25}{2}\Big[\,1 + 2(1.28403) + 2(1.64872) + 2(2.117) + 2.71828\,\Big]$$

Step 9. Do each multiplication inside the bracket and add the results:

$$\text{bracket} = \ 1 + 2.56806 + 3.29744 + 4.234 + 2.71828 \ = \ 13.81778$$

Step 10. Multiply by the front factor:

$$T_4 = 0.125 \times 13.81778 = 1.72722.$$

Step 11. Compute M_4.

Step 12. Formula used (Midpoint Rule, $n = 4$):

$$M_n = \Delta x\,\Big[f(\bar{x}_1) + f(\bar{x}_2) + \cdots + f(\bar{x}_n)\Big].$$

For $n = 4$ this is

$$M_4 = \Delta x\Big[f(\bar{x}_1) + f(\bar{x}_2) + f(\bar{x}_3) + f(\bar{x}_4)\Big].$$

Step 13. Width of each subinterval:

$$\Delta x = \frac{b-a}{n} = \frac{1-0}{4} = 0.25.$$

Step 14. Midpoints of the subintervals, $\bar{x}_i = \dfrac{x_{i-1} + x_i}{2}$:

$$\bar{x}_1 = \frac{0 + 0.25}{2} = 0.125 \qquad \bar{x}_2 = \frac{0.25 + 0.5}{2} = 0.375$$
$$\bar{x}_3 = \frac{0.5 + 0.75}{2} = 0.625 \qquad \bar{x}_4 = \frac{0.75 + 1}{2} = 0.875$$

Step 15. Function values at the midpoints (rounded to 6 significant figures where necessary):

$$f(\bar{x}_1) = e^{0.125} = 1.13315 \qquad f(\bar{x}_2) = e^{0.375} = 1.45499$$
$$f(\bar{x}_3) = e^{0.625} = 1.86825 \qquad f(\bar{x}_4) = e^{0.875} = 2.39888$$

Step 16. Add the 4 midpoint values (all weights are 1):

$$\text{bracket} = 1.13315 + 1.45499 + 1.86825 + 2.39888 = 6.85527$$

Step 17. Multiply by Δx:

$$M_4 = 0.25 \times 6.85527 = 1.71382.$$

Step 18. Conclusion:

$$1.71382 \le \int_0^1 e^x \, dx \le 1.72722,$$

so the integral lies in an interval of length 0.0134 found without any antiderivative.

Step 19. Check against the exact value $\displaystyle\int_0^1 e^x \, dx = e - 1 = 1.71828$: indeed $1.71382 < 1.71828 < 1.72722$. Check against the error bounds with $K = e$: $E_T \le \dfrac{e}{12 \cdot 16} = 0.01416$ and $E_M \le \dfrac{e}{24 \cdot 16} = 0.007079$; the actual errors 0.00894 and 0.00446 satisfy them.

Answer. $M_4 \approx 1.7138 \le \displaystyle\int_0^1 e^x dx \le T_4 \approx 1.7272$ (T_4 over, M_4 under).

Practice Problems

Midpoint and Trapezoidal Rules

1. Approximate $\displaystyle\int_0^4 \sqrt{x}\,dx$ using T_4 (grid points $0, 1, 2, 3, 4$; use $\sqrt{0} = 0,\ \sqrt{1} = 1,\ \sqrt{2} \approx 1.414,\ \sqrt{3} \approx 1.732,\ \sqrt{4} = 2$).

Your working:

2. Approximate the same integral, $\displaystyle\int_0^4 \sqrt{x}\,dx$, using M_4 (midpoints $0.5, 1.5, 2.5, 3.5$; use $\sqrt{0.5} \approx 0.707,\ \sqrt{1.5} \approx 1.225,\ \sqrt{2.5} \approx 1.581,\ \sqrt{3.5} \approx 1.871$).

Your working:

3. Approximate $\displaystyle\int_0^{\pi/2} \cos x\, dx$ using T_4 and M_4 (so $\Delta x = \frac{\pi}{8}$). Compare both with the exact value, and say whether each rule over- or under-estimates and why.

Your working:

4. The integral $\int_0^1 e^{-x^2}\,dx$ has no elementary antiderivative. Approximate it using T_4, M_4 and S_4, and compare each with the value 0.746824 obtained by computer.

Your working:

5. Approximate the area of the half-disc $\int_{-1}^{1} \sqrt{1-x^2}\,dx$ using T_4 and M_4. Compare with the exact area $\frac{\pi}{2}$, explain the signs of the errors using f'', and explain why the error-bound theorem cannot be used for this integral.

Your working:

Simpson's Rule

6. Approximate $\displaystyle\int_0^2 (x^3+1)\,dx$ using S_4 (grid points $0, 0.5, 1, 1.5, 2$), and compare to the exact value. Explain why the two must match exactly.

Your working:

7. Approximate $\displaystyle\int_0^{\pi} \sin x\, dx$ using S_4 (grid points $0, \frac{\pi}{4}, \frac{\pi}{2}, \frac{3\pi}{4}, \pi$; use $\sin 0 = 0,\ \sin\frac{\pi}{4} \approx 0.707,\ \sin\frac{\pi}{2} = 1,\ \sin\frac{3\pi}{4} \approx 0.707,\ \sin\pi = 0$), and compare to the exact value 2.

Your working:

8. Approximate $\displaystyle\int_1^2 \frac{1}{x}\,dx$ using Simpson's Rule S_4 and compare with the exact value $\ln 2$.

Your working:

9. Use Simpson's Rule with $n = 2$ to approximate $\displaystyle\int_{-1}^{3} (x^3 - 2x^2 + 4x - 1)\,dx$. Compute the exact value and explain the result.

Your working:

10. Let $f(x) = x^4$ on $[0, 2]$. (a) Compute S_2 and its exact error. (b) Compute S_4 and its exact error. (c) Compare each error with the Simpson error bound and find the ratio of the two errors.

Your working:

Numerical integration from tables of data

11. A car accelerates from rest and its speed $v(t)$ (m/s) is measured every 2 seconds:

t (s)	0	2	4	6	8	10	12
$v(t)$	0	6.6	11.0	14.0	16.0	17.3	18.2

Estimate the distance travelled in the first 12 seconds, $\displaystyle\int_0^{12} v(t)\,dt$, using (a) T_6, (b) S_6, (c) M_3 (with $n = 3$ subintervals of width 4 s).

Your working:

12. The depth $d(x)$ (metres) of a river was measured every 10 m across its width:

x (m)	0	10	20	30	40	50	60	70	80
$d(x)$	0	2.1	3.4	4.2	4.6	4.1	3.3	1.8	0

Using the table: (a) estimate the cross-sectional area $\displaystyle\int_0^{80} d(x)\,dx$ using T_8; (b) estimate it using S_8; (c) estimate it using M_4 (subintervals of width 20 m); (d) if the water flows at an average speed of 0.8 m/s, estimate the discharge (area $\times$ speed) using your Simpson area.

Your working:

Error bounds

13. Find a bound on the error in approximating $\int_0^1 e^x \, dx$ using T_8.

Your working:

14. Find a bound on the error in approximating $\int_1^3 \ln x \, dx$ using M_6.

Your working:

15. Tasks: (a) find a bound on the error in approximating $\displaystyle\int_1^4 \sqrt{x}\,dx$ using T_6; (b) compute T_6 and the exact value, and check that the actual error respects your bound.

Your working:

16. Find a bound on the error in approximating $\int_1^2 \frac{1}{x}\,dx$ using S_4, and compare it with the actual error found in Problem 8.

Your working:

Choosing n for a required accuracy

17. How large must n be so that T_n approximates $\displaystyle\int_0^1 e^x\,dx$ with error less than 0.001?

Your working:

18. How large must (even) n be so that S_n approximates $\int_0^2 x^4\,dx$ with error less than 0.01?

Your working:

19. How large must n be so that M_n approximates $\displaystyle\int_0^2 e^{-x^2}\,dx$ with error less than 10^{-4}? (Find K by maximising $|f''|$ on $[0,2]$.)

Your working:

20. How large must (even) n be so that Simpson's Rule S_n approximates $\int_1^3 \ln x \, dx$ with error less than 10^{-6}?

Your working:

21. For $\displaystyle\int_0^1 e^x\,dx$, find the smallest n that guarantees an error below 10^{-6} for (a) the Trapezoidal Rule T_n and (b) Simpson's Rule S_n (even n). Compare the number of function evaluations ($n+1$ in each case).

Your working:

Concavity, comparison of the rules and exact values

22. Without computing any sum, use the sign of f'' to decide whether T_n and M_n overestimate or underestimate each integral. Then confirm with T_4 and M_4. (a) $\displaystyle\int_1^3 \frac{1}{x}\,dx$. (b) $\displaystyle\int_1^5 \ln x\,dx$. (c) $\displaystyle\int_0^2 x^3\,dx$. (d) $\displaystyle\int_0^\pi \sin x\,dx$.

Your working:

23. A student computes T_4 and M_4 for the same integral and gets $T_4 = 5.20$ and $M_4 = 5.05$. Without further calculation, give the best available estimate for the Simpson value with 8 subintervals using the identity $S_{2n} = \dfrac{2M_n + T_n}{3}$, and explain why this weighted average reduces the error compared with either T_4 or M_4 alone.

Your working:

24. For $\displaystyle\int_1^2 \frac{1}{x}\,dx$, compute T_2 and M_2, then use $S_{2n} = \dfrac{2M_n + T_n}{3}$ to obtain S_4. Compare with the value of S_4 computed directly in Problem 8.

Your working:

25. Explain, using the error-bound formulas, why doubling n reduces the Trapezoidal error bound by a factor of 4 but reduces the Simpson error bound by a factor of 16.

Your working:

26. For $\displaystyle\int_1^2 \frac{dx}{x}$, compute T_2, T_4, S_2, S_4, the actual error of each against $\ln 2$, and the ratios $\dfrac{|I - T_2|}{|I - T_4|}$ and $\dfrac{|I - S_2|}{|I - S_4|}$. Compare them with the factors 4 and 16 predicted by the error bounds.

Your working:

27. The integral $\displaystyle\int_0^1 \frac{4}{1+x^2}\,dx$ equals π. Approximate it using T_4, M_4 and S_4, compute the actual error of each, and rank the three rules from most to least accurate.

Your working:

28. Let $f(x) = \ln x$ on $[1, 3]$ with $n = 4$. Without computing, put L_4, R_4, T_4, M_4 and the exact integral I in increasing order, justifying each inequality from the monotonicity and concavity of f; then confirm numerically. (L_4 and R_4 are the left- and right-endpoint Riemann sums.)

Your working:

Error diagnosis

29. A student approximates $\displaystyle\int_0^2 x^2\,dx$ with S_4 and writes

$$S_4 = \frac{0.5}{3}\big[f(0) + 2f(0.5) + 4f(1) + 2f(1.5) + f(2)\big] = \frac{13}{6} \approx 2.167.$$

The student says that since Simpson's Rule is exact for polynomials of degree at most 3, the rule must be wrong for this problem. Find the error and give the correct result.

Your working:

30. To make the Midpoint Rule M_n approximate $\displaystyle\int_0^1 e^x\,dx$ with error less than 0.001, a student writes

$$\frac{e\cdot 1^3}{12n^2} < 0.001 \implies n^2 > 226.5 \implies n = 16.$$

Find the error and give the correct smallest n.

Your working:

31. A student claims: "For a positive function, the Trapezoidal Rule always overestimates the integral, because the trapezoids stick out above the curve." Find the error by computing T_4 for $\displaystyle\int_0^{\pi} \sin x \, dx$, and state the correct rule.

Your working: